Problem 3: Verity
5 Points

Problem ID: `verity`
Rank: `1`

Introduction

Hey, it's me, it's Big Bennity! Don't ask me anything! There is a new trending game called verity. Fearing of not being able to keep up with the Gen Alpha brainrotity, Big Bennity obviously participates in a round of guessing the verity.



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Problem Statement

This is an interactive problem! Please make sure to read the Interaction Format section before attempting this problem.

In this problem, your task is to discover a hidden string through a series of interactions with the judge.

Initially, you are given the length of the hidden string $\mathbf{N}$ and a list of $\mathbf{K}$ characters $\mathbf{A_1}, \mathbf{A_2}, \ldots, \mathbf{A_K}$ denoting the possible characters that can be used in the hidden string.

Then, your program can make up to 100 guesses. For each guess:

- If your guess is incorrect, the judge will respond with the hidden string but with all the incorrect positions of your guess replaced with _ (an underscore).
- If your guess is correct, the judge will respond with `CORRECT` and your program proceeds to the next test case.

For example, if the hidden string was `OBESITY` and we inputted the guess `VARIETY`, the judge will respond with `_____TY`

*Note: Templates are available for this problem—and **all other problems in this contest**—in Python, Java, and C++! Find them in the contest.zip provided at the start of the contest. Templates handle input and output for you, so you can just fill out a single function!*

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Interaction Format

This is an interactive problem! Unlike regular problems, your program and the judge will run simultaneously. Please see the contest guide for more information. Please flush your buffer as instructed by this post when you output, or use our template code that handles it for you. If you run into technical issues with interaction, please let us know with a clarification request! The first line of the input contains a single integer $\mathbf{T}$ denoting the number of test cases.

For each test case:

- The first line contains two integers $\mathbf{N}$, denoting the length of the sequence, and $\mathbf{K}$, denoting the number of types of -ities.
- The second line contains $\mathbf{K}$ space-separated uppercase characters $\mathbf{A_1}$ $\mathbf{A_2}$ ... $\mathbf{A_K}$, representing the names of the -ities.

Then, for each test case:

1. Output a single line containing the string representing your guess. This string must consist only of characters in $\mathbf{A}$.
2. Receive a single line of input. You will receive one of two results:
   (a) The string `CORRECT`, indicating that you have guessed correctly. This test case will end and the next one (if there is one) will begin.
   (b) A string consisting of at least 1 and at most $\mathbf{N}$ _ (underscore) characters. The remaining characters which are not underscores will match the characters submitted in your most recent guess, indicating that you guessed the correct character for that position.
3. Repeat the first two steps until you receive the string `CORRECT`, at which point you will move onto the next test case.

Constraints

$1 \leq \mathbf{T} \leq 100$
$1 \leq \mathbf{N} \leq 100$
$1 \leq \mathbf{K} \leq 26$
Every $\mathbf{A}_i$ for $1 \leq i \leq \mathbf{K}$ will be an uppercase letter `ABCDEFGHIJKLMNOPQRSTUVWXYZ`.

The number of guesses you submit per test case must be less than **100**.

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Sample Interaction

Sample Input Download

```
2
4 4
L C F V
L___
L__V
CORRECT
10 5
A B C D E
__AB_____D_
E_AB_E__DA
ECAB_EA_DA
ECAB_EACDA
CORRECT
```

Sample Output Download

```
LVVF
LCFV
LFCV
AAABCCCEDD
EBABDEDBDA
ECABBEADDA
ECABAEACDA
ECABEEACDA
```

Sample Explanations

The judge begins by outputting `2`, the number of test cases. **Note that the guesses made in this interaction may not be optimal, and are simply sample guesses. Also note that the answer strings you see in this sample interaction are not guaranteed to be the actual answer strings provided in the sample test cases: your program must be able to guess an arbitrary answer string, which is randomly generated by the judge**.

For the first test case, the answer string contains 4 characters, which are all either `L`, `F`, `C`, or `V`. Suppose we begin by guessing the string `LVVF`. The judge outputs the string `L___`, indicating that we correctly guessed that `L` falls in the first position, but that we guessed incorrectly for the remaining positions. We guess twice more, and on the 3rd guess (`LFCV`), the judge outputs `CORRECT`, indicating that we successfully guessed the string.

The second test case is similar, but the answer string contains 10 characters, which are each one of `A`, `B`, `C`, `D`, or `E`.

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